Business Question and Visualization Report

Date	27 October 2026
Team ID	Individual
Project Name	EV Market Performance & Sustainability Analytics
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

This report defines business questions for the EV Market Performance & Sustainability Analytics project. The questions are answered using Tableau worksheets and dashboards built from the supplied EV market dataset. The visualisations focus on sales, manufacturing countries, vehicle categories, battery technologies, range, charging and efficiency, manufacturer performance, and reported CO_2 savings.

Business Questions and Visualisations

1. How do annual EV sales vary by model year and vehicle category?
    - Visualisation: Multi-line chart showing annual sales by model year and category.
    - Screenshot of visualisation: Insert a screenshot of the corresponding Tableau worksheet.
2. 2. Which manufacturing countries have the highest recorded EV sales?
    - Visualisation: Horizontal bar chart comparing sales by manufacturing country.
    - Screenshot of visualisation: Insert a screenshot of the corresponding Tableau worksheet.
3. 3. How are EV sales distributed across vehicle categories?
    - Visualisation: Donut chart showing sales contribution by vehicle category.
    - Screenshot of visualisation: Insert a screenshot of the corresponding Tableau worksheet.

Note: The report below provides more than the required eight business questions and maps each question to an existing Tableau visualisation.

Business Questions, Visualisations and Interpretation

4. Which battery technologies have the largest sales shares?

Visualisation: Donut chart, EV Sales by Battery Technology.

Interpretation: The supplied dashboard screenshot shows NMC at approximately 33.80%, LFP at 31.99%, NCA at 17.23%, and Sodium-Ion at 9.88%; the remaining share is Solid-State. These percentages describe the displayed dataset.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

5. What is the overall EV market profile?

Visualisation: Four KPI cards for total sales, average vehicle price, average driving range, and annual CO_2 saved.

Interpretation: The screenshot displays approximately 1.55B total sales units, $64,921 average vehicle price, 324.54 km average driving range, and 737.93K tonnes annual CO_2 savings. Confirm aggregation and units in the source before formal reporting.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

6. How does driving range vary across vehicle categories?

Visualisation: Range Distribution chart.

Interpretation: Compare the spread and central values for the vehicle categories. Use Tableau tooltips to inspect individual values and avoid drawing conclusions from chart shape alone.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

7. Which country/category combinations show higher or lower sales?

Visualisation: EV Sales Heatmap by Country and Category.

Interpretation: Colour intensity makes differences between manufacturing countries and vehicle categories easier to scan and helps identify combinations for further investigation.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

8. What relationship appears between charging power and energy efficiency?

Visualisation: Charging Speed vs Energy Efficiency scatter plot.

Interpretation: The scatter plot compares maximum DC charging power with the efficiency measure and uses mark size for battery capacity. The displayed axis values look unusually large, so check aggregation and units before making numerical claims.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

9. How do manufacturer sales and customer-rating measures compare?

Visualisation: Manufacturer Performance bubble chart.

Interpretation: The chart is intended to compare manufacturers across sales and the customer-rating measure. The displayed rating axis appears inconsistent with a typical rating scale, so validate the field and aggregation before ranking manufacturers.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

10. How do sales trends change across model years?

Visualisation: Sales Trend line chart, with the Model Year filter where configured.

Interpretation: Use the year filter to focus on a period and compare the lines. Treat the chart as a historical view, not a forecast.

Screenshot of visualisation: Insert the matching worksheet/dashboard screenshot here.

Evidence note: The interpretations above use the supplied Tableau dashboard screenshots. The underlying dataset was not independently recalculated for this report, so values should be checked against the workbook before submission.